



Designing a Racing Bicycle Using Computational Software

Members: Nicholas, Farris, Kaleab, Johnny

Introduction

- Objective: Enhancing speed, agility, and aerodynamics for competitive racing.
- Designing a high-performance racing bicycle using computational software.
 - Ansys Workbench



WORKBENCH

Literature Review



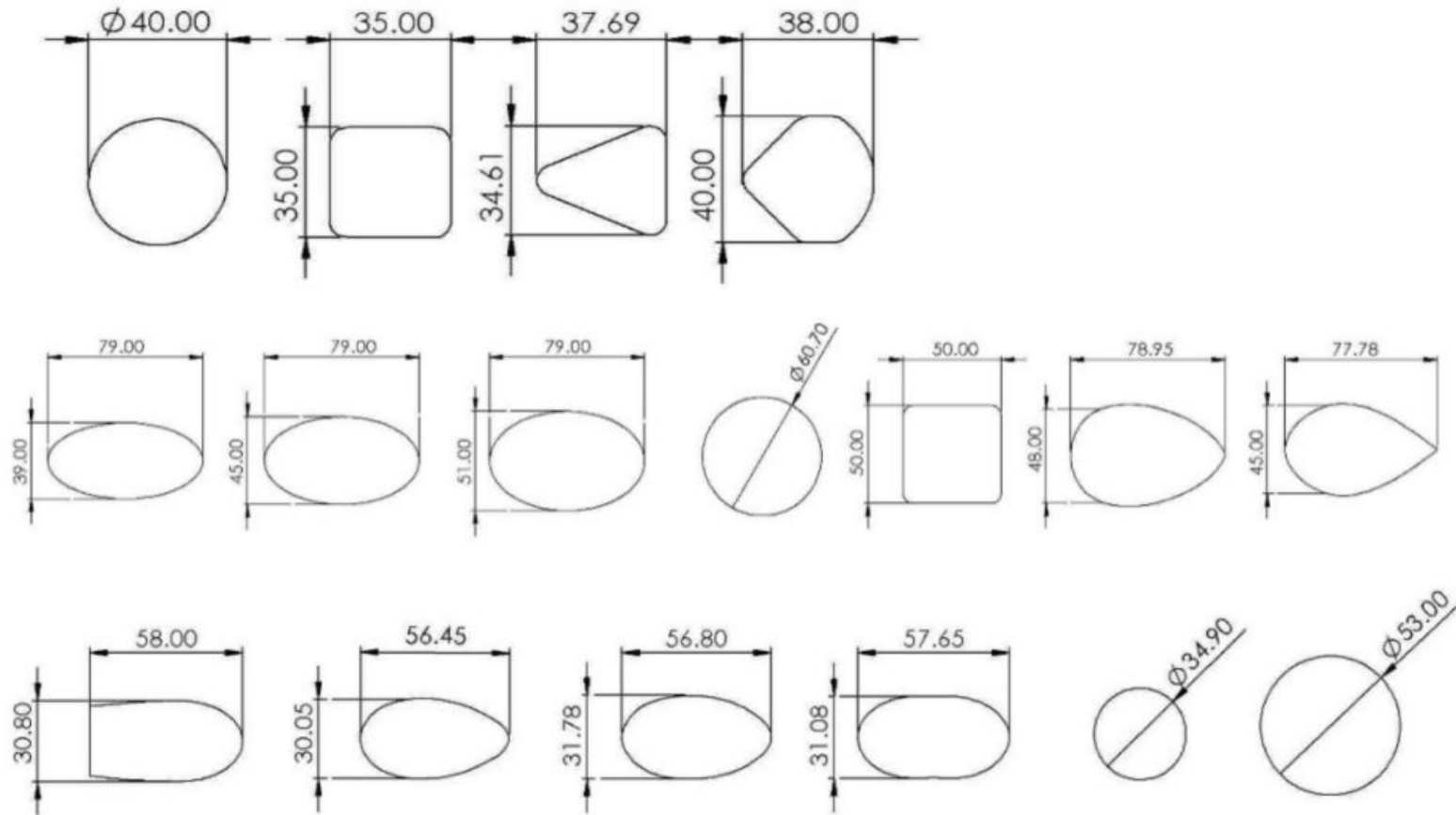
Materials	Modulus of Elasticity (GPa)	Poisson's Ratio	Density (Kg/M3)
Steel	205	0.29	7800
Aluminum 6061 T6	72	0.33	2700
Titanium Grade 9	95	0.30	4480
CFRP	415	0.10	1800

Material Selection



Bike Components	Option 1	Option 2	Option 3
Frame	HSD	LSD	HSLRD
Material	Aluminum	Carbon Fiber	Steel
Transmission System	Chain	Toothed Belts	Carbon Shaft
Wheel Tire	Mountain Tire	Gravel Tire	Road Tire

Cross Sectional Study



Frame Parts	Cross-Section	Displacement	Mass	Weighted Drag	Cost
Top Tube	cylindrical	✓	✓	✓	✓
	rectangular	✓	x	x	✓
	triangular	x	✓	✓	x
Down Tube	oval	✓	✓	✓✓	✓
	cylindrical	x	✓	x	✓
	rectangular	x	✓	x	✓
Seat Tube	Cylindrical (53 mm)	✓	x	x	✓
	Cylindrical (35 mm)	x x	✓	✓	✓
Seat Stays	Oval (17x31 mm)	✓	x	x	✓
	rectangular	✓✓	x	✓	x
	cylindrical	x	✓✓	x	✓
Chain Stays	Rectangular (20 x 60 mm)	✓✓	x x	x x	✓
	Cylindrical	x x x	✓✓	✓✓	✓

- ✓ indicates a good behavior–result
- ✓✓ indicates a very good behavior–result
- x indicates a bad behavior–result
- x x indicates a very bad behavior–result
- x x x indicates an unacceptable result.

Design Process Overview



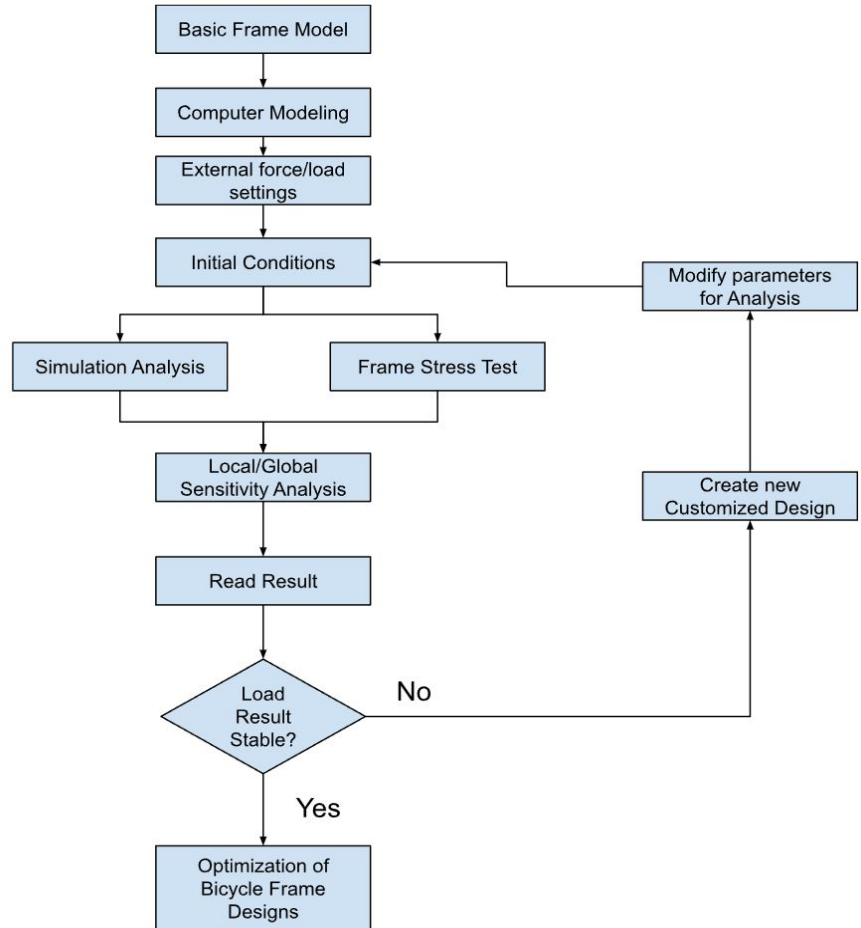
- Pre-defined Loading Conditions for analysis.

Loading Conditions	Description
Static Startup	Load of 700 N and 200 N applied to the top of the seat tube and on the bottom of the bracket
Steady State Pedaling	Load of 1000 N and 200 N applied to the bottom tube and head tube respectively
Vertical Impact	Load of 2250 N applied to the top of the bracket
Horizontal Impact	Load of 2250 N applied to the head tube horizontally
Rear Wheel Braking	Load of 750 N applied to the rear wheel bracket

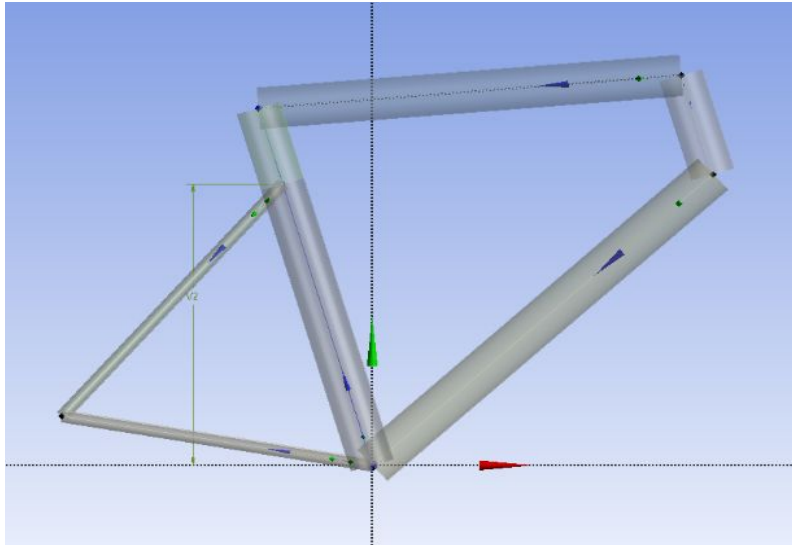


Iterative Optimization

- Bike frame optimization flowchart.



Prototype Development



Conclusion



- Unique bicycle frame design with varying cross sections
 - Particular attention to the design of the bicycle frame cross sections
- Improvement in Stiffness-to-weight ratio,
- Pedalling efficiency
- Frame Design
 - Upright position of a high seat frame → comfortable
 - High seats often allow for a more relaxed riding posture → reduce strain on back
 - Lower Rear end → maximum power transfer and efficiency



Thank You for Listening!